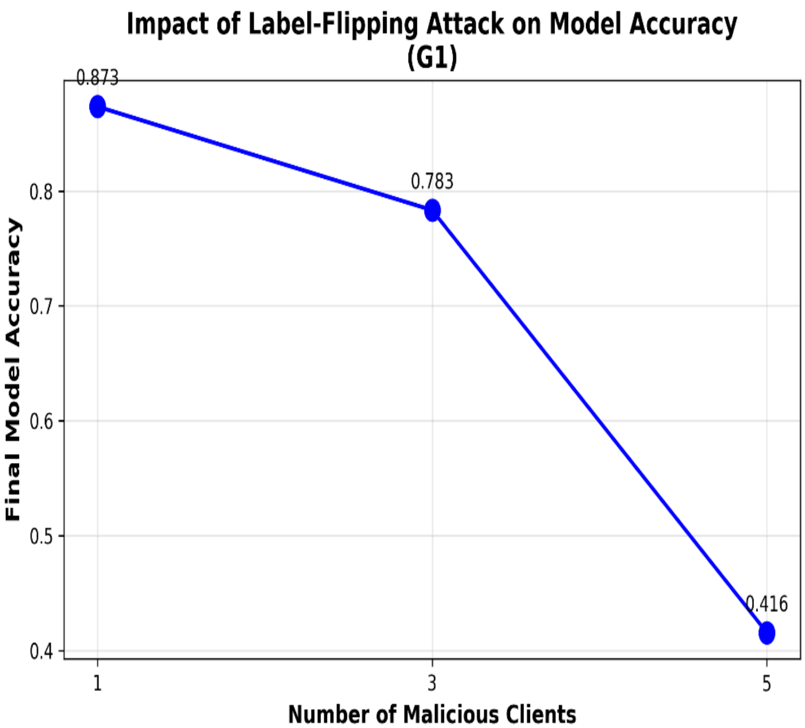
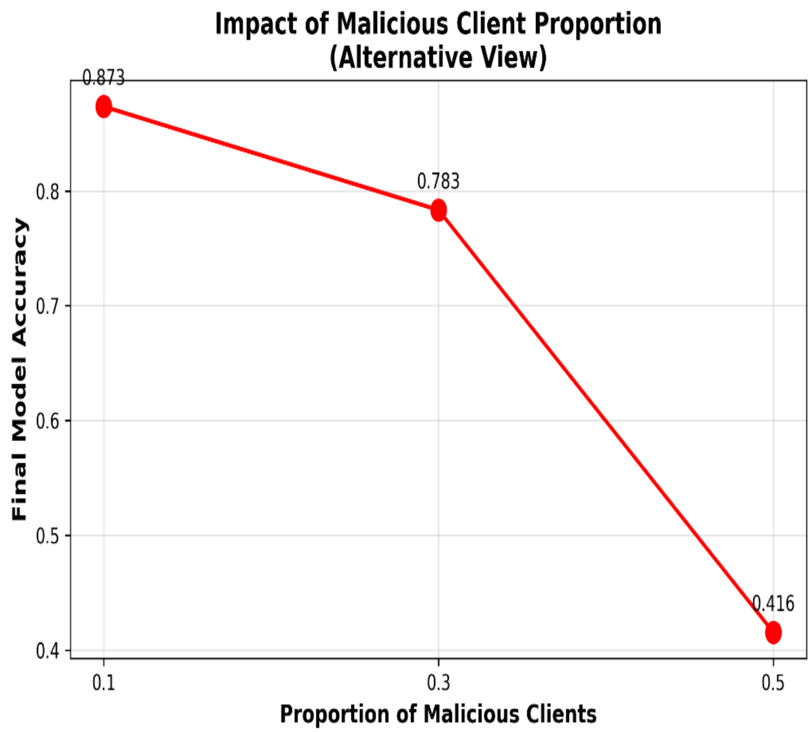
Ex1: Implementation of label-flipping attacks

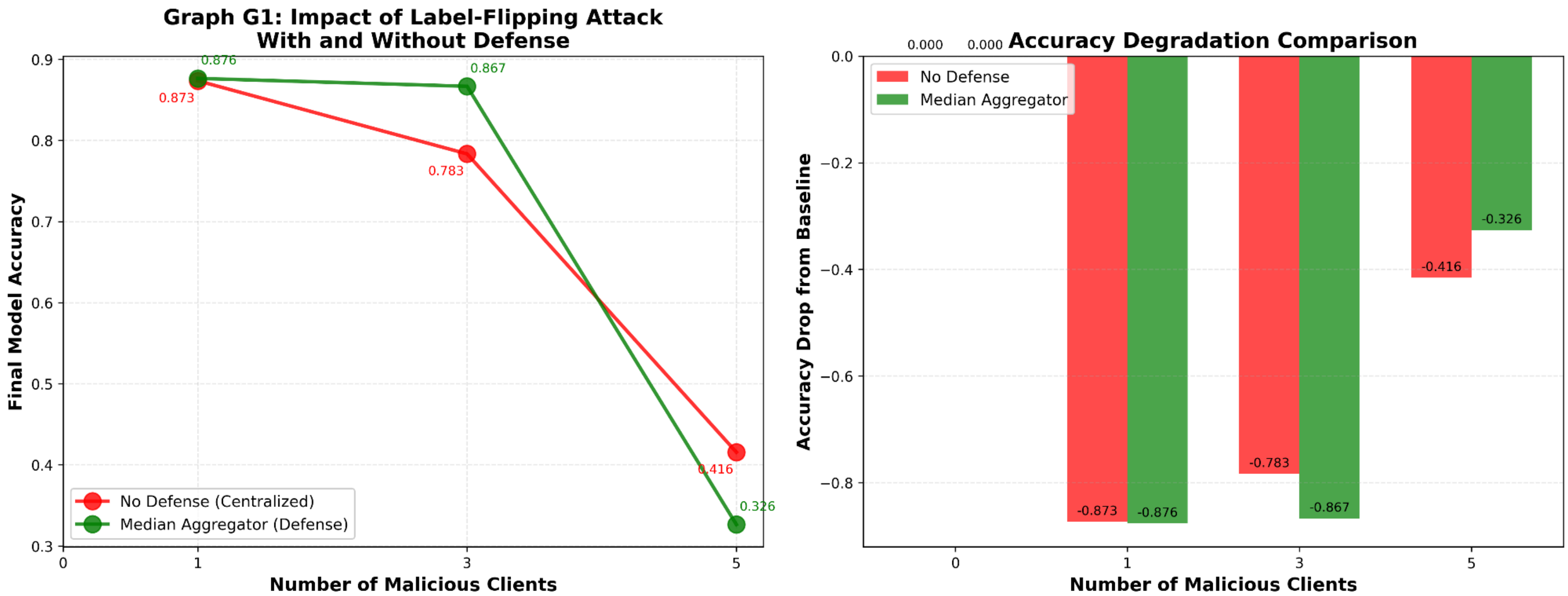
G1 graph clearly shows that model accuracy declines as the proportion of malicious clients increases. With only one malicious client (10%), the attack has minimal effect, and the model retains high accuracy. However, once the proportion reaches 30%, there is a noticeable drop in performance, indicating that the aggregated gradient begins to be corrupted. At 50% malicious clients, the model accuracy collapses to nearly 40%, demonstrating that the label-flipping attack severely disrupts FedAvg when adversaries become a significant fraction of the total clients. These results confirm that FedAvg is **not resilient** to label-flipping and that its vulnerability grows proportionally with the number of malicious clients.

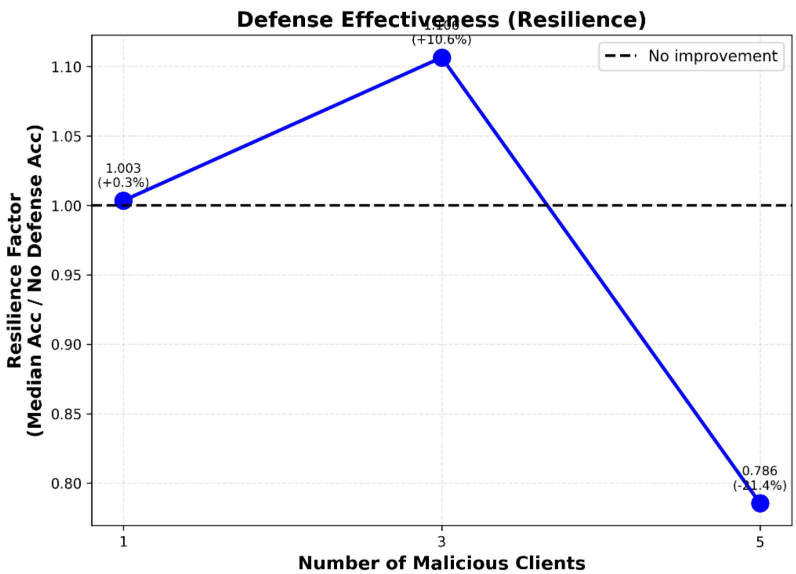




Ex2: Implementation of Defenses

Using the **median aggregator** significantly improves performance when the proportion of malicious clients increases. For **prop = 0.1 and 0.3**, the median defense protects the model by reducing the influence of malicious updates, resulting in much higher accuracy than the standard FedAvg. However, when **50% of the clients are malicious**, the defense becomes less effective because the attackers represent half of the updates, and even a coordinate-wise median can be pulled toward the flipped labels. Overall, the median aggregator provides strong robustness for low–to–moderate attack strengths, but it cannot fully resist extreme poisoning where half the clients are adversaries.





Summary Table: Accuracy Comparison

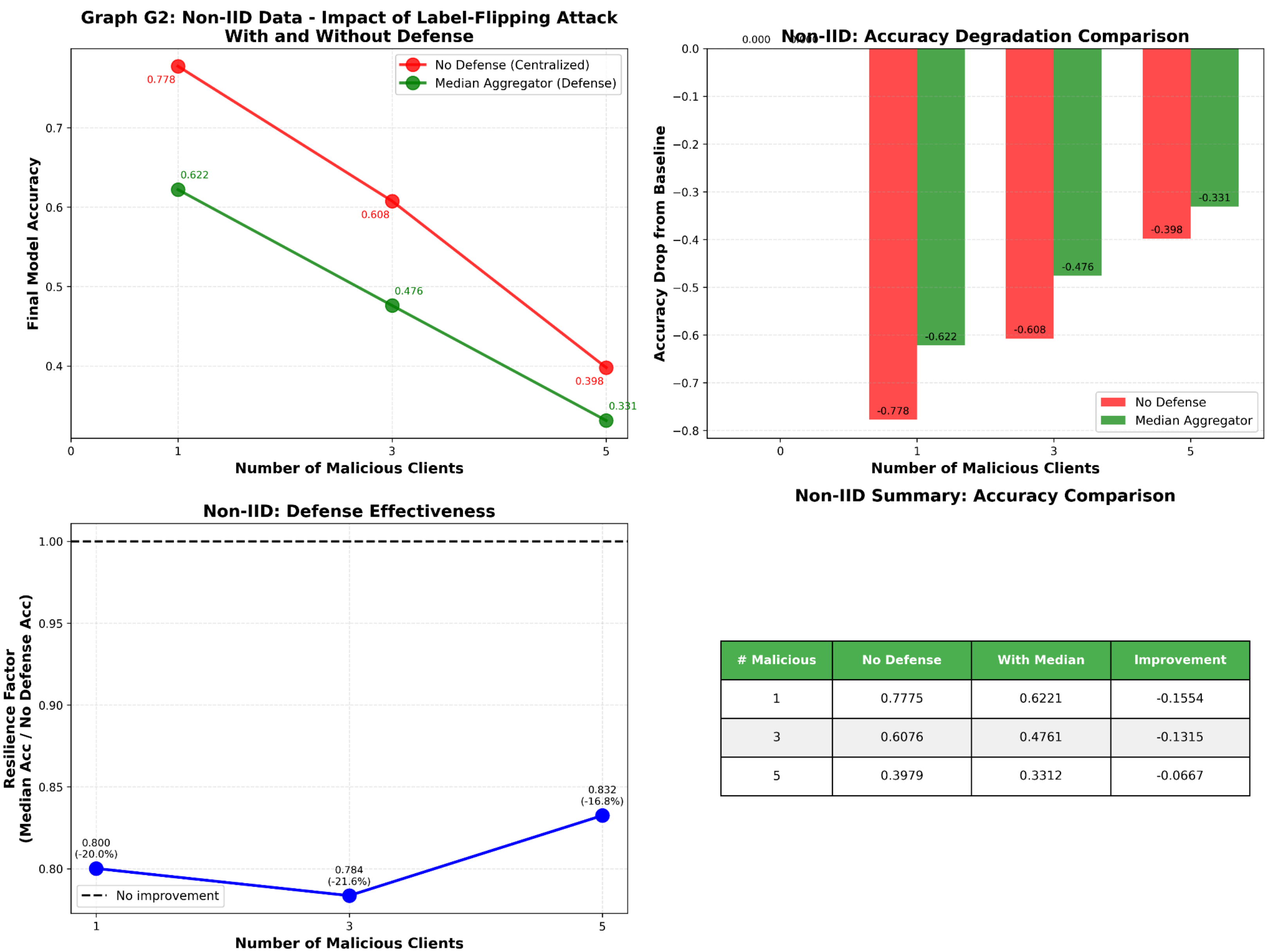
# Malicious	No Defense	With Median	Improvement
1	0.8735	0.8764	+0.0030
3	0.7834	0.8667	+0.0833
5	0.4155	0.3264	-0.0891

Ex3: Simulation of non-iid case

In the non-IID scenario, both the baseline FedAvg and the median aggregator perform worse than in IID settings. The non-IID distribution causes client gradients to diverge significantly, making the global model more unstable even without attacks. As the proportion of malicious clients increases, model accuracy decreases for both methods.

However, unlike in the IID case, the median defense no longer improves robustness. In fact, the median aggregator consistently performs worse than the centralized FedAvg. This occurs because median aggregation assumes that benign updates are similar and that malicious updates appear as outliers. In non-IID settings, this assumption breaks down since benign updates are highly diverse. As a result, the median cannot correctly isolate malicious updates and ends up degrading model convergence.

Overall, while the median defense is effective in IID settings, it becomes counter-productive under non-IID data distributions.



# Malicious	No Defense	With Median	Improvement
1	0.7775	0.6221	-0.1554
3	0.6076	0.4761	-0.1315
5	0.3979	0.3312	-0.0667